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15.4 Photoperiodism

Trees take several years for initiation of flowering whereas an annual herb flowers within few months. Each plant requires a specific time period to complete their vegetative phase which will be followed by reproductive phase as per their internal control points through Biological Clock. The physiological mechanisms in relation to flowering are controlled by (i) light period (Photoperiodism) and (ii) temperature (Vernalization). The physiological change on flowering due to relative length of light and darkness (photoperiod) is called **Photoperiodism**. The term photoperiodism was coined by **Garner** and **Allard** (1920) when they observed this in 'Biloxi' variety of soybean (*Glycine max*) and 'Maryland mammoth' variety of tobacco (*Nicotiana tabacum*). The photoperiod required to induce flowering is called **critical day length**. Maryland mammoth (tobacco variety) requires 12 hours of light and cocklebur (*Xanthium pensylvanicum*) requires 15.05 hours of light for flowering.

1. Classification of plants based on Photoperiodism

Depending upon the photoperiodic responses plants are classified as given in Figure 15.24.

i. **Long day plants**: The plants that require long critical day length for flowering are called long day plants or short night plants. Example: Pea, Barley and Oats.

ii. **Short long day plants**: These are long day plants but should be exposed to short day lengths during early period of growth for flowering. Example: Wheat and Rye.

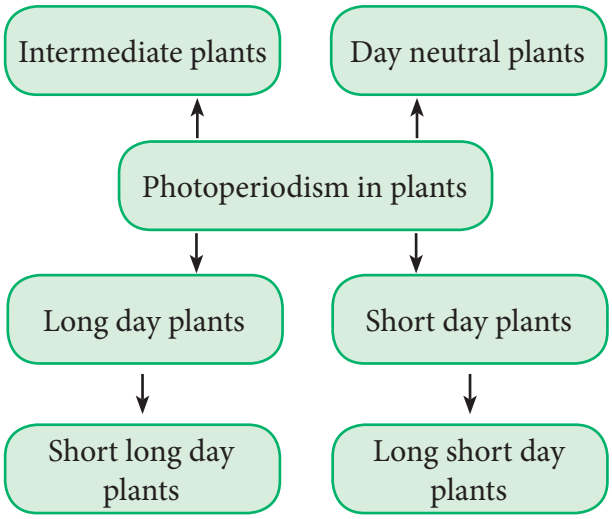

Figure: 15.24 Classification of Plants based on Photoperiodism

iii. **Short day plants**: The plants that require a short critical day length for flowering are called short day plants or long night plants. Example: Tobacco, Cocklebur, Soybean, Rice and *Chrysanthemum*.

iv. **Long short day plants**: These are actually short-day plants but they have to be exposed to long days during their early periods of growth for flowering. Example: Some species of *Bryophyllum* and Night jasmine.

v. **Intermediate day plants:** These require a photoperiod between long day and short day for flowering. Example: Sugarcane and *Coleus*.

vi. **Day neutral plants**: There are a number of plants which can flower in all possible photoperiods. They are also called **photo neutrals** or **indeterminate plants**. Example: Potato, *Rhododendron*, Tomato and Cotton.

2. Photoperiodic induction

An appropriate photoperiod in 24 hours' cycle constitutes one inductive cycle. Plants may require one or more inductive cycles for flowering. The phenomenon of